1. Explain what **X** and **y** represent.

Answer:

- X represents the independent variable (house size in m²).

- **y** represents the dependent variable (house price in thousands).

2. What do theta_0 and theta_1 represent in the regression equation?

Answer:

(theta_0) = intercept (price when house size = 0).

(theta_1) = slope (price increase per 1 m²).

3.Is the prediction reasonable based on the dataset? Why?

Answer:

No, the prediction is not reasonable because the model diverged due to a large learning rate. The parameters θ_0 and θ_1 became extremely large, which resulted in unrealistic predictions far from the dataset values.

4. Explain:

o Why SSE decreases over time

Answer:

Because Gradient Descent updates parameters to minimize prediction error.

o What convergence means in Gradient Descent

Answer:

Convergence means that the algorithm has reached optimal parameter values where the error no longer decreases significantly. At this point, Gradient Descent has found the best-fitting line for the data.

5.What happens if the learning rate is too large?

Answer:

If the learning rate is too large, Gradient Descent may diverge instead of converging. The algorithm overshoots the minimum of the cost function, causing the parameters to oscillate or explode and the error (SSE) to increase instead of decrease.